Abdul janeeb k

'at' is used for schedule one time tasks.

- For this task we want to install 'at'
- Sudo Yum install -y at
- After that need to start and enable.
- Sudo systemctl start atd
- Sudo systemctl enable atd
- Sudo systemctl status atd
- Suppose we want to list scheduled jobs we use atq this will show scheduled tasks.
- Atq (it will show pending tasks)
- For remove a task using command is atrm jobnumber . eg:- atrm 1
- Syntax
- at now + 2 minutes (it means execute tasks after 2 minutes)
- at 5 pm aug 15 2026 (this like syntax also we can use)
- at -c job ID (displays the execution script of a scheduled job.)

```
[root@Astra ~]# yum install at
Last metadata expiration check: 0:24:53 ago on Thu 15 Oct 2026 09:56:36 AM IST.
Dependencies resolved.
================================================================================
 Package         Architecture        Version                     Repository           Size
================================================================================
Installing:
 at              x86_64              3.1.23-13.el9_7             baseos               60 k

Transaction Summary
================================================================================
Install  1 Package

Total download size: 60 k
Installed size: 123 k
Is this ok [y/N]: y
Downloading Packages:
at-3.1.23-13.el9_7.x86_64.rpm                                  125 kB/s |  60 kB     00:00
--------------------------------------------------------------------------------
Total                                                           47 kB/s |  60 kB     00:01
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Installing       : at-3.1.23-13.el9_7.x86_64                              1/1
  Running scriptlet: at-3.1.23-13.el9_7.x86_64                              1/1
Created symlink /etc/systemd/system/multi-user.target.wants/atd.service → /usr/lib/systemd/sy
stem/atd.service.

  Verifying        : at-3.1.23-13.el9_7.x86_64                              1/1

Installed:
  at-3.1.23-13.el9_7.x86_64

Complete!
[root@Astra ~]# systemctl status atd
o atd.service - Deferred execution scheduler
     Loaded: loaded (/usr/lib/systemd/system/atd.service; enabled; preset: enabled)
     Active: inactive (dead) since Thu 2026-10-15 10:20:09 IST; 1min 52s ago
   Duration: 23min 2.590s
       Docs: man:atd(8)
   Main PID: 13096 (code=exited, status=0/SUCCESS)
        CPU: 23ms

Oct 15 09:57:07 Astra systemd[1]: Started Deferred execution scheduler.
Oct 15 09:59:00 Astra atd[13119]: Starting job 1 (a0000101c7bd2d) for user 'root' (0)
Oct 15 09:59:00 Astra atd[13119]: pam_unix(atd:session): session opened for user root(uid=0) by root(uid=0)
```

```
[root@Astra ~]# systemctl status atd
○ atd.service - Deferred execution scheduler
     Loaded: loaded (/usr/lib/systemd/system/atd.service; enabled; preset: enabled)
     Active: inactive (dead) since Thu 2026-10-15 10:20:09 IST; 1min 52s ago
   Duration: 23min 2.590s
       Docs: man:atd(8)
   Main PID: 13096 (code=exited, status=0/SUCCESS)
        CPU: 23ms

Oct 15 09:57:07 Astra systemd[1]: Started Deferred execution scheduler.
Oct 15 09:59:00 Astra atd[13119]: Starting job 1 (a0000101c7bd2d) for user 'root' (0)
Oct 15 09:59:00 Astra atd[13119]: pam_unix(atd:session): session opened for user root(uid=0) by root(uid=0)
Oct 15 09:59:00 Astra atd[13119]: pam_unix(atd:session): session closed for user root
Oct 15 10:03:00 Astra atd[13161]: Starting job 2 (a0000201c7bd31) for user 'root' (0)
Oct 15 10:03:00 Astra atd[13161]: pam_unix(atd:session): session opened for user root(uid=0) by root(uid=0)
Oct 15 10:03:00 Astra atd[13161]: pam_unix(atd:session): session closed for user root
Oct 15 10:20:09 Astra systemd[1]: Stopping Deferred execution scheduler...
Oct 15 10:20:09 Astra systemd[1]: atd.service: Deactivated successfully.
Oct 15 10:20:09 Astra systemd[1]: Stopped Deferred execution scheduler.
[root@Astra ~]# systemctl start atd
[root@Astra ~]# systemctl enable atd
[root@Astra ~]# systemctl status atd
● atd.service - Deferred execution scheduler
     Loaded: loaded (/usr/lib/systemd/system/atd.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-10-15 10:22:10 IST; 1min 10s ago
       Docs: man:atd(8)
   Main PID: 13677 (atd)
      Tasks: 1 (limit: 7621)
     Memory: 284.0K (peak: 400.0K)
        CPU: 3ms
     CGroup: /system.slice/atd.service
             └─13677 /usr/sbin/atd -f

Oct 15 10:22:10 Astra systemd[1]: Started Deferred execution scheduler.
```

```
[root@Astra ~]# ls
grep-practice  janeeb.txt
[root@Astra ~]# cat janeeb.txt
i am a linux admin
[root@Astra ~]# datetimectl
-bash: datetimectl: command not found
[root@Astra ~]# timedatectl
               Local time: Thu 2026-10-15 10:30:24 IST
           Universal time: Thu 2026-10-15 05:00:24 UTC
                 RTC time: Thu 2026-10-15 05:00:25
                Time zone: Asia/Kolkata (IST, +0530)
System clock synchronized: no
              NTP service: active
          RTC in local TZ: no
[root@Astra ~]# at 5 pm oct 15 2026
warning: commands will be executed using /bin/sh
at> echo " this is a test " > 5pm.txt<EOT>
job 4 at Thu Oct 15 17:00:00 2026
[root@Astra ~]# at 5 pm oct 15 2026
warning: commands will be executed using /bin/sh
at> touch b<EOT>
job 5 at Thu Oct 15 17:00:00 2026
[root@Astra ~]# atq
4       Thu Oct 15 17:00:00 2026 a root
5       Thu Oct 15 17:00:00 2026 a root
[root@Astra ~]# atrm 5
[root@Astra ~]# atq
4       Thu Oct 15 17:00:00 2026 a root
```

```
4       Thu Oct 15 17:00:00 2020 a job
[root@Astra ~]# at -c 4
#!/bin/sh
# atrun uid=0 gid=0
# mail root 0
umask 22
SHELL=/bin/bash; export SHELL
HISTCONTROL=ignoredups; export HISTCONTROL
HISTSIZE=1000; export HISTSIZE
HOSTNAME=Astra; export HOSTNAME
PWD=/root; export PWD
LOGNAME=root; export LOGNAME
XDG_SESSION_TYPE=tty; export XDG_SESSION_TYPE
MOTD_SHOWN=pam; export MOTD_SHOWN
HOME=/root; export HOME
LANG=en_US.UTF-8; export LANG
LS_COLORS=rs=0:di=01\;34:ln=01\;36:mh=00:pi=40\;33:so=01\;35:do=01\;35:bd=40\;33\;01:cd=40\;3
3\;01:or=40\;31\;01:mi=01\;37\;41:su=37\;41:sg=30\;43:ca=30\;41:tw=30\;42:ow=34\;42:st=37\;44
:ex=01\;32:\*.tar=01\;31:\*.tgz=01\;31:\*.arc=01\;31:\*.arj=01\;31:\*.taz=01\;31:\*.lha=01\;3
1:\*.lz4=01\;31:\*.lzh=01\;31:\*.lzma=01\;31:\*.tlz=01\;31:\*.txz=01\;31:\*.tzo=01\;31:\*.t7z
=01\;31:\*.zip=01\;31:\*.z=01\;31:\*.dz=01\;31:\*.gz=01\;31:\*.lrz=01\;31:\*.lz=01\;31:\*.lzo
=01\;31:\*.xz=01\;31:\*.zst=01\;31:\*.tzst=01\;31:\*.bz2=01\;31:\*.bz=01\;31:\*.tbz=01\;31:\*
.tbz2=01\;31:\*.tz=01\;31:\*.deb=01\;31:\*.rpm=01\;31:\*.jar=01\;31:\*.war=01\;31:\*.ear=01\;
31:\*.sar=01\;31:\*.rar=01\;31:\*.alz=01\;31:\*.ace=01\;31:\*.zoo=01\;31:\*.cpio=01\;31:\*.7z
=01\;31:\*.rz=01\;31:\*.cab=01\;31:\*.wim=01\;31:\*.swm=01\;31:\*.dwm=01\;31:\*.esd=01\;31:\*
.jpg=01\;35:\*.jpeg=01\;35:\*.mjpg=01\;35:\*.mjpeg=01\;35:\*.gif=01\;35:\*.bmp=01\;35:\*.pbm=
01\;35:\*.pgm=01\;35:\*.ppm=01\;35:\*.tga=01\;35:\*.xbm=01\;35:\*.xpm=01\;35:\*.tif=01\;35:\*
.tiff=01\;35:\*.png=01\;35:\*.svg=01\;35:\*.svgz=01\;35:\*.mng=01\;35:\*.pcx=01\;35:\*.mov=01
\;35:\*.mpg=01\;35:\*.mpeg=01\;35:\*.m2v=01\;35:\*.mkv=01\;35:\*.webm=01\;35:\*.webp=01\;35:\
*.ogm=01\;35:\*.mp4=01\;35:\*.m4v=01\;35:\*.mp4v=01\;35:\*.vob=01\;35:\*.qt=01\;35:\*.nuv=01\
;35:\*.wmv=01\;35:\*.asf=01\;35:\*.rm=01\;35:\*.rmvb=01\;35:\*.flc=01\;35:\*.avi=01\;35:\*.fl
i=01\;35:\*.flv=01\;35:\*.gl=01\;35:\*.dl=01\;35:\*.xcf=01\;35:\*.xwd=01\;35:\*.yuv=01\;35:\*
.cgm=01\;35:\*.emf=01\;35:\*.ogv=01\;35:\*.ogx=01\;35:\*.aac=01\;36:\*.au=01\;36:\*.flac=01\;
36:\*.m4a=01\;36:\*.mid=01\;36:\*.midi=01\;36:\*.mka=01\;36:\*.mp3=01\;36:\*.mpc=01\;36:\*.og
g=01\;36:\*.ra=01\;36:\*.wav=01\;36:\*.oga=01\;36:\*.opus=01\;36:\*.spx=01\;36:\*.xspf=01\;36
:; export LS_COLORS
SSH_CONNECTION=192.168.1.18\ 53828\ 192.168.1.158\ 22; export SSH_CONNECTION
XDG_SESSION_CLASS=user; export XDG_SESSION_CLASS
SELINUX_ROLE_REQUESTED=; export SELINUX_ROLE_REQUESTED
LESSOPEN=\|\|/usr/bin/lesspipe.sh\ %s; export LESSOPEN
USER=root; export USER
SELINUX_USE_CURRENT_RANGE=; export SELINUX_USE_CURRENT_RANGE
SHLVL=1; export SHLVL
XDG_SESSION_ID=3; export XDG_SESSION_ID
XDG_RUNTIME_DIR=/run/user/0; export XDG_RUNTIME_DIR
SSH_CLIENT=192.168.1.18\ 53828\ 22; export SSH_CLIENT
which_declare=declare\ -f; export which_declare
```

```
; export LS_COLORS
SSH_CONNECTION=192.168.1.18\ 53828\ 192.168.1.158\ 22; export SSH_CONNECTION
XDG_SESSION_CLASS=user; export XDG_SESSION_CLASS
SELINUX_ROLE_REQUESTED=; export SELINUX_ROLE_REQUESTED
LESSOPEN=\|\|/usr/bin/lesspipe.sh\ %s; export LESSOPEN
USER=root; export USER
SELINUX_USE_CURRENT_RANGE=; export SELINUX_USE_CURRENT_RANGE
SHLVL=1; export SHLVL
XDG_SESSION_ID=3; export XDG_SESSION_ID
XDG_RUNTIME_DIR=/run/user/0; export XDG_RUNTIME_DIR
SSH_CLIENT=192.168.1.18\ 53828\ 22; export SSH_CLIENT
which_declare=declare\ -f; export which_declare
PATH=/root/.local/bin:/root/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin; export PAT
H
SELINUX_LEVEL_REQUESTED=; export SELINUX_LEVEL_REQUESTED
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/0/bus; export DBUS_SESSION_BUS_ADDRESS
MAIL=/var/spool/mail/root; export MAIL
SSH_TTY=/dev/pts/0; export SSH_TTY
cd /root || {
        echo 'Execution directory inaccessible' >&2
        exit 1
}
${SHELL:-/bin/sh} << 'marcinDELIMITER0b0b1479'
echo " this is a test " > 5pm.txt
marcinDELIMITER0b0b1479
```